Digital Twin-Assisted Retail Theft Prevention

1. Papers Reviewed

1. Agentic Incident Response through Digital Twin-Enhanced Multiscale Planning (2026)
2. Hierarchical Agentic Incident Response with Digital-Twin-Validated Attack Inference (2026)
3. From Offline to Periodic Adaptation for Pose-Based Shoplifting Detection in Real-World Retail Security (2026)
4. Suspicious Behavior Detection with Temporal Feature Extraction and Time-Series Classification (2023)
5. Criminal Intention Detection at Early Stages of Shoplifting Cases by Using 3DCNN (2021)
6. Hybrid Digital Twin Solutions for Real-Time Threat Prevention in AI-Driven IoT Networks (2025)
7. Anomaly Detection in Industrial IoT Networks via Hybrid Digital Twins and Continual Learning (2026)
8. A Survey on Digital Twins: Architecture, Enabling Technologies, Security and Privacy (2023)
9. Linking Physical Objects to Their Digital Twins via Invisible Fiducial Markers (2021)
10. Leveraging Digital Twin Technologies for Public Space Protection and Vulnerability Assessment (2024)

2. Main Findings from the Papers

Paper 1 – Digital twin for validating actions

Gao et al. studied digital twins for autonomous incident response. Their system uses a digital twin as a safe environment where an AI agent can test commands before applying them to the real system. The study used a virtual five-server network and a DeepSeek-R1-Distill-Qwen-14B model fine-tuned with LoRA. The system achieved more than 90% recovery success and reduced recovery time compared with earlier agent-based methods.

The useful idea for the proposed theft prevention work is the validation step. Instead of allowing an automated decision to directly affect the real environment, the action can first be checked in the digital twin. This idea can be transferred to retail by testing a predicted theft event or response in the virtual store.

Paper 2 – Checking an inferred attack using a digital twin

The second paper also uses a digital twin, but focuses on correcting the AI's understanding of an incident. The system first infers an attack sequence from alerts, replays it in a 33-component virtual network, and compares the result with real observations. This improved attack-action inference from about 78% to 98.7%. Recovery success was between 86% and 90% in the tested scenarios.

For theft prevention, the important point is that the digital twin can be used to check whether a suspected event is actually consistent with the available evidence. This can help reduce false alarms before an alert is raised.

Paper 3 – Shoplifting detection that adapts over time

Yao et al. focused directly on shoplifting detection in real retail environments. Instead of keeping raw video, the system extracts 17 body keypoints and uses models such as STG-NF, SPARTA and TSGAD. It periodically updates the models using new store data. Their RetailS dataset contains around 20 million normal frames, 898 staged theft events and 53 real incidents.

The adaptive approach performed better than static offline models in 91.6% of the evaluations. The models could also run at about 26 FPS on edge hardware. A limitation is that hidden hands or objects can make pose-based detection difficult. This paper is especially useful for the sensing part of the proposed system because it supports privacy-friendly and lightweight behaviour detection.

Paper 4 – Detecting suspicious movement using simple features

Nazir et al. converted video into person trajectories using YOLOv5m and DeepSort. Instead of processing the complete video with a heavy 3D model, they used the movement information as a time series and tested models such as MiniRocket and XceptionTime.

Using the shoplifting subset of UCF-Crime, the method was reported to be 8.45 times faster than RTFM. MiniRocket and XceptionTime reached a median F1-score of about 92%. However, tracking can become difficult when people overlap, and the method does not capture detailed hand movements.

The main lesson is that theft-related behaviour does not always require full video processing. Simpler movement features can be faster and may be suitable for edge processing.

Paper 5 – Detecting behaviour before theft happens

Martínez-Mascorro et al. tried to detect suspicious behaviour before the actual theft. They used a four-layer 3DCNN and divided videos into pre-crime, suspicious behaviour and crime evidence segments. The network was trained using the pre-crime segment so that it learned behaviour occurring before the theft.

The reported accuracy was around 75%, while confirmatory tests reached 100% recall for suspicious behaviour. The study also showed that large 3D video inputs require high computation. Manual segmentation of pre-crime behaviour is another limitation.

This paper is important for the proposed work because it supports the idea of predicting possible theft before the item is actually taken.

Paper 6 – Digital twin with real-time threat response

Thiyagarajan et al. proposed a hybrid digital twin for AI-based IoT security. The system connects physical devices with a virtual representation and uses machine learning to detect abnormal activity. It also includes an automated response mechanism and federated learning for privacy.

The reported detection accuracy was 95%, with an average mitigation time of 0.4 seconds. The system was tested with simulated networks containing up to 10,000 devices. The main drawback is the computation needed to continuously maintain the digital twin.

For retail, this paper supports the idea of combining real-time sensing with a virtual store model and using the model to support quick decisions.

Paper 7 – Continual learning with a hybrid digital twin

Melis et al. studied anomaly detection in industrial IoT systems where the normal behaviour can change over time. Their architecture combines a virtual environment with physical hardware and uses continual learning to update the detection model.

The system maintained 98.34% accuracy and 98.23% F1-score. The authors also showed that using a compact 9 MB feature representation could produce results close to retraining on 1 GB of raw data, while reducing training time from about 720 seconds to 45 seconds.

This is useful for retail because store behaviour can also change when layouts, customers or operating conditions change. A system that can update itself would be more practical than a fixed model.

Paper 8 – Digital twin architecture and security

Wang et al. provided a broad survey of digital twins and their security and privacy issues. The paper describes the relationship between the physical entity, its digital twin and the systems used to exchange information between them. It also discusses threats such as privacy problems, model poisoning, communication attacks and synchronization issues.

Although this paper does not focus on theft detection, it helps in designing the overall digital twin architecture. It also shows that security and privacy should be considered as part of the system rather than added later.

Paper 9 – Linking real objects with digital twins

Schwartz et al. proposed invisible markers that can be used to connect a physical object to its digital twin. The markers use optical properties that are not normally visible and can act as a unique physical fingerprint.

The markers were reported to be readable from several metres away and stable up to 200°C. The approach needs special cameras, filters and light sources, so it is not immediately suitable for a normal retail store.

For the proposed system, this paper is more of an optional extension. Similar item-level identification could help connect valuable products with their digital representations.

Paper 10 – Digital twins for protection of public spaces

Stefanidou et al. studied the use of digital twins for protecting public spaces. Their architecture combines information from CCTV, LiDAR, RF receivers, sonar, microphones and other sources. The digital environment represents people and assets and can be used for event detection and vulnerability assessment.

The study considered several complex attack scenarios. Its main limitation is the high computational requirement and the privacy issues caused by large-scale tracking.

The useful idea for retail is the layered architecture. A store can similarly have a real-world sensing layer, a digital representation layer and an analysis layer that identifies possible threats.

3. Comparison of the Main Ideas

Paper	Main idea	Useful for theft prevention	Main limitation
1	Digital twin checks AI actions	Safe validation before response	Needs repeated simulation
2	Digital twin checks inferred events	Can reduce false alarms	Slow for complex scenarios
3	Adaptive pose-based shoplifting detection	Direct retail application; privacy-friendly	Hands/objects may be hidden
4	Movement-based suspicious behaviour detection	Fast and lightweight detection	Limited fine-grained actions
5	Pre-crime behaviour detection	Supports early theft prediction	Manual video segmentation
6	Hybrid DT with real-time response	Links sensing and response	High computation
7	Continual learning with DT	Adapts to changing conditions	Needs careful update strategy
8	DT architecture and security survey	Helps design secure DT	No direct theft experiment
9	Physical item to DT linking	Possible item-level tracking	Special hardware required
10	Multi-sensor DT for protection	Useful overall architecture	High cost and privacy concerns

4. Learning

The papers show that theft prevention can be approached in three connected parts. First, the system needs a way to observe behaviour. Pose estimation and movement-based methods are useful because they can reduce the need to process and store complete video.

Second, the system should be able to represent the store and its current state. The digital twin papers show that a virtual representation can combine information from different sources and maintain a model of what is happening.

Third, the digital twin can be used for checking and improving decisions. Papers 1 and 2 show that an AI-generated decision can be tested or replayed in a virtual environment before it is trusted. Papers 3 and 7 also show the importance of adapting the detection system as conditions change.

5. Research Gap

Most of the reviewed papers solve only one part of the problem. Some concentrate on detecting shoplifting from video, while others focus on digital twins for cyber or industrial security. The direct combination of early theft detection, a retail digital twin, continuous updating and decision validation is not clearly addressed in the reviewed work.

Another important gap is practical retail deployment. A real store has changing layouts, different customer behaviour, occlusion between people, privacy requirements and limited edge computing resources. Therefore, a useful system should not depend only on heavy video models. It should combine lightweight sensing with a digital representation of the store and use the twin to support better decisions.

6. Conclusion

The reviewed literature shows that digital twins are useful not only for visualization but also for monitoring, simulation and decision checking. The shoplifting studies provide methods for identifying suspicious and pre-crime behaviour, while the digital twin studies provide ideas for synchronization, validation and adaptation.

The main opportunity is to combine these ideas for retail theft prevention. A system that detects behaviour early, maintains a digital representation of the store and checks suspicious events before raising an alert could provide a more practical approach than using video detection alone.

References – Papers Reviewed

1. Gao, Y., Li, T., Hammar, K. “Agentic Incident Response through Digital Twin-Enhanced Multiscale Planning.” 2026.

2. Gao, Y., Chen, J., Li, T. “Hierarchical Agentic Incident Response with Digital-Twin-Validated Attack Inference.” 2026.
3. Yao, S., Rashvand, N., Danesh Pazho, A., Tabkhi, H. “From Offline to Periodic Adaptation for Pose-Based Shoplifting Detection in Real-World Retail Security.” IEEE Internet of Things Journal, 2026.
4. Nazir, A., Mitra, R., Sulieman, H., Kamalov, F. “Suspicious Behavior Detection with Temporal Feature Extraction and Time-Series Classification.” Sensors, 2023.
5. Martínez-Mascorro, G. A. et al. “Criminal Intention Detection at Early Stages of Shoplifting Cases by Using 3DCNN.” Computation, 2021.
6. Thiyagarajan, R., Mohana Geetha, D., Mudassar Ali, A., Sreekanth, K. “Hybrid Digital Twin Solutions for Real-Time Threat Prevention in AI-Driven IoT Networks.” IJCESEN, 2025.
7. Melis, A., Piroddi, A., Girau, R. et al. “Anomaly Detection in Industrial IoT Networks via Hybrid Digital Twins and Continual Learning.” Internet of Things, 2026.
8. Wang, Y., Su, Z., Luan, T. H. et al. “A Survey on Digital Twins: Architecture, Enabling Technologies, Security and Privacy.” IEEE Communications Surveys & Tutorials, 2023.
9. Schwartz, M., Geng, Y., Lenzini, G., Lagerwall, J. P. F. “Linking Physical Objects to Their Digital Twins via Invisible Fiducial Markers.” 2021.
10. Stefanidou, A., Varlamis, I., Papadopoulos, G. Th. et al. “Leveraging Digital Twin Technologies for Public Space Protection and Vulnerability Assessment.” 2024.